Level 1 Functional Skills Mathematics
Sample 4

Duration: 25 minutes
Total marks: 15 marks

City & Guilds
A City & Guilds Group Business

Version: 1.0

MODEL SOLUTION SCORE
15 / 15 marks (100% Pass)
25 minutes · Verified by MFS

SECTION 1 – CALCULATOR NOT PERMITTED

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

Please check that your name is correctly printed on the candidate barcode label. If not, please tell the invigilator before the start of the exam.

You should have the following for this assessment:

- a pen with black or blue ink
- a pencil
- an eraser
- a 30cm ruler.



You must NOT use a protractor.
You must NOT use a calculator for Section 1.

General instructions

- Read through each question carefully.
- You may use a dictionary.
- Write all your answers in this booklet.
- Check your calculations and check that your answers make sense.

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SECTION 1 – CALCULATOR NOT PERMITTED

There are **15** marks available in this section.

You should check all your work as you go along.

You must **not** use a calculator in this section.



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Q1

(4 + 2) x 6 – 1 =

• BIDMAS: Brackets (4 + 2) = 6 --> Multiply 6 x 6 = 36 --> 36 – 1 = 35

35

_____ [1 mark] _____ **(1 mark)**

Q2

What is 0.4 as a percentage?

• 0.4 x 100% = 40% (multiply decimal by 100)

40

_____% [1 mark] **(1 mark}**

Q3

What is 8.35719 rounded to two decimal places?

• 8.35719: 3rd decimal place is 7 (≥ 5), so round up 5 to 6 --> 8.36

8.36

_____ [1 mark] **(1 mark)**

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Q4

4.978 x _____ = 4978 1 000

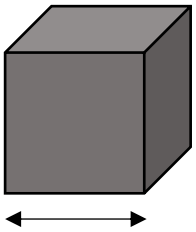
What is the missing number?

- 4,978 ÷ 4.978 = 1,000 (decimal point moved 3 places right)

[1 mark]

(1 mark)

Q5



5cm

What is the volume of this cube?

- Volume of Cube = length × width × height = 5 × 5 × 5 = 125 cm³

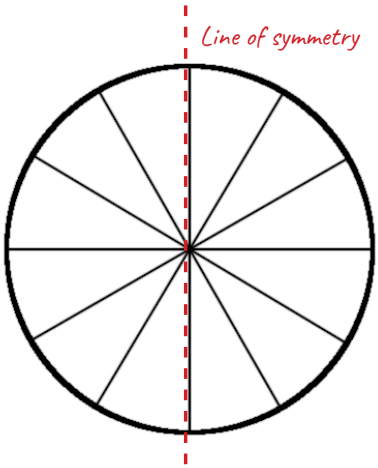
125 _____ cm^3

[1 mark]

(1 mark)

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Q6



Shade $\frac{2}{3}$ of the circle so there is at least one line of symmetry.

- Circle has 12 equal sectors: 2/3 of 12 = (12 ÷ 3) × 2 = 8 sectors to shade
- Shading 4 sectors on left and 4 on right creates a vertical line of symmetry [A1]

[1 mark]

Q7

Which of these lists of fractions and mixed numbers is in order from smallest to largest?

A $2\frac{1}{4}$ $1\frac{3}{4}$ $\frac{3}{4}$ $1\frac{1}{4}$ ☐

B $1\frac{1}{4}$ $\frac{3}{4}$ $1\frac{3}{4}$ $2\frac{1}{4}$ ☐

- Decimals: ¾ = 0.75, 1 ¼ = 1.25, 1 ¾ = 1.75, 2 ¼ = 2.25
- 0.75 < 1.25 < 1.75 < 2.25
  --> List C is in order [A1]

C $\frac{3}{4}$ $1\frac{1}{4}$ $1\frac{3}{4}$ $2\frac{1}{4}$ ☑

D $\frac{3}{4}$ $1\frac{3}{4}$ $1\frac{1}{4}$ $2\frac{1}{4}$ ☐

[1 mark]

(1 mark)

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Q8

If 28 x 63 = 1764, what is 1764 ÷ 63?

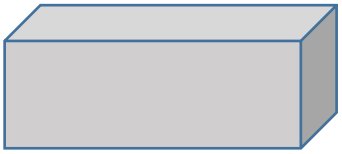
- Since 28 x 63 = 1,764, then 1,764 ÷ 63 = 28 (inverse calculation)

28

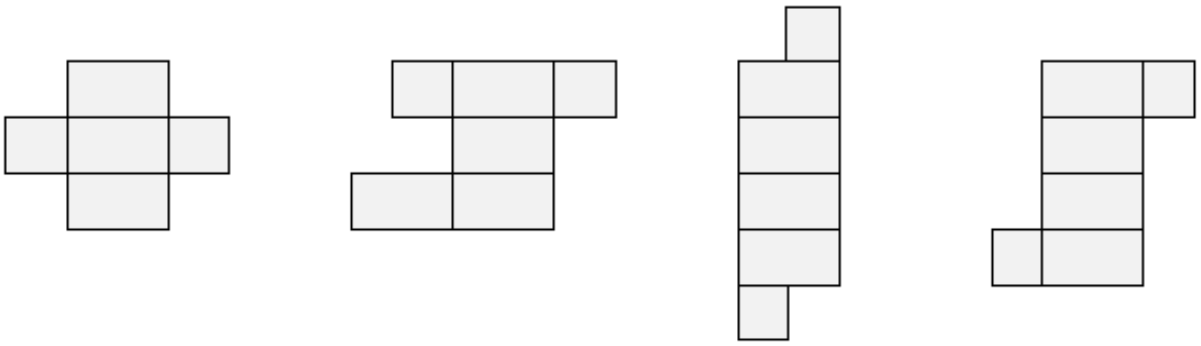
[1 mark]

(1 mark)

Q9



Which one of these nets will fold into this shape?



A **B** **C** **D**

- Shape is a triangular prism with 2 triangular faces and 3 rectangular faces
- Net D correctly connects the 3 rectangles with 2 opposite triangles [A1]

D

[1 mark]

(1 mark)

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Q10

Write two hundred and five thousand and thirty in figures.

- 205 thousands = 205 000, plus thirty = 205 030

205 030

[1 mark]

(1mark)

Q11

A shop owner sells a new board game on a website.

He wants to give an average (mean) customer rating for his website.

He asks customers to give a star rating for the board game.

Star Ratings – 8 customers							
3	4.5	3.5	3	5	0.5	4	4.5

What rating will he use for the website?

Show your working

- Calculate Mean Rating = Total Stars ÷ Number of Customers
- Total = 3 + 4.5 + 3.5 + 3 + 5 + 0.5 + 4 + 4.5 = 28 stars [M1]
- Mean = 28 ÷ 8 = 3.5 stars [A1]

3.5

[2 marks]

(2 marks)

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Q12

A baker wants to order enough flour for 10 loaves of bread weighing 750g each.

She has a recipe for a 500g loaf of bread which needs 480g of flour.

How many kilograms of flour does the baker need?

Show your working

Step 1: Find scale factor for 750g loaf from 500g recipe:

750 g ÷ 500 g = 1.5 times larger [M1]

Step 2: Flour for 1 loaf = 480 g × 1.5 = 720 g flour [M1]

Step 3: Flour for 10 loaves = 10 × 720 g = 7,200 g [M1]

Step 4: Convert to kilograms = 7,200 ÷ 1,000 = 7.2 kg [A1]

7.2

[3 marks]

_____ kg

(3 marks)

End of Section 1

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Level 1 Functional Skills Mathematics
Sample 4

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Version: 1.1

Duration: 1 hour 20 minutes
Total marks: 45 marks

MODEL SOLUTION SCORE
45 / 45 marks (100% Pass)
1 hr 20 mins · Verified by MFS

SECTION 2 – CALCULATOR PERMITTED

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

• If you have used any additional answer sheets write the number of additional sheets in this box. ☐
• Please ensure that you **staple** additional answer sheets to the **back** of this booklet, clearly labelling them with your full name, enrolment number, centre number and date in BLOCK CAPITALS.
• You must use a black or blue pen. You may use a pencil for charts and diagrams.
***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

Please check that your name is correctly printed on the candidate barcode label. If not, please tell the invigilator before the start of the exam.

You should have the following for this assessment

- a pen with black or blue ink.
- a pencil
- an eraser
- a 30cm ruler
- a calculator.





You must NOT use a protractor.

General instructions

- Read through each question carefully.
- You may use a dictionary.
- Show your working out (where required).
- Write all your working out and answers in this booklet.
- Check your calculations and check that your answers make sense.
- There are additional pages at the back of this booklet if you run out of space or ask the invigilator if you need additional sheets of paper.

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SECTION 2 – CALCULATOR PERMITTED

There are **45** marks in this section.

You should check all your work as you go along.

You may use a calculator.



Q1

What is 0.35 as a fraction? Give your answer in its simplest form.

• 0.35 = 35/100 = (35÷5) / (100÷5) = 7/20 (divide by 5)

7/20

[1 mark]

$\frac{\square}{\square}$

(1 mark)

Q2

A stadium has capacity for 28 000 people to watch sporting events.

21 500 people attended the last event held at the stadium.

Approximately, how full was the stadium?

(tick one box)

A $\frac{2}{3}$ full ☐

B $\frac{4}{5}$ full ☐

C $\frac{3}{4}$ full ☑

D $\frac{9}{10}$ full ☐

• 21,500 ÷ 28,000 ≈ 21,000 ÷ 28,000 = 3/4 full (Option C)

[1 mark]

(1 mark)

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Q3

The range of temperature recorded in London one day is 8ºC.

Which pair of temperatures could have been the minimum and maximum that day?

(tick one box)

A 10ºC and 16ºC ☐

B 8ºC and 10ºC ☐

C -3ºC and 4ºC ☐

D -2ºC and 6ºC ☑

[1 mark] **(1 mark)**

- Lowest temperature = −2°C, Highest = 6°C (Difference = 8°C)

Q4

74035.52 ÷ 3.7 =

- 74,035.52 ÷ 3.7 = 20,009.6

20 009.6

[1 mark] **(1 mark)**

Q5

What is $\frac{4}{5}$ of 690?

- 4/5 of 690 = (690 ÷ 5) × 4 = 138 × 4 = 552

552

[1 mark] **(1 mark)**

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Q6

A salesperson must drive from Bristol to attend a meeting in Nottingham at 10am.

This table shows distances in miles between different places.

City	Bristol	Nottingham	Derby
Bristol		140	135
Nottingham	140		15
Derby	135	15	

He works out he needs to leave Bristol at 9am.

Is his calculation sensible?
Explain your answer.

Is his calculation sensible? *(tick one box)* **Yes** ☐ **No** ☐

Explanation

No, because travelling 140 miles in 1 hour would require an average speed of 140 mph. This is far too fast and greatly exceeds UK road speed limits (70 mph).

(1 mark)

[1 mark]

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Q7

A woman wants to buy new kitchen units for £1 800

The shop offers some plans for the customer to make equal **monthly** payments.

Monthly Payment Plans	
Number of years	Total interest charged
1	5%
2	15%
3	25%

The woman wants to know how much her **monthly** payments will be if she pays for the units over **two** years.

What will the **monthly** payments be?

Show your working

Step 1: Interest for 2 years = 15% of £1,800: £1,800 × 0.15 = £270 [M1]

Step 2: Total amount to pay = £1,800 + £270 = £2,070 [A1]

Step 3: Months in 2 years = 2 × 12 = 24 months

Step 4: Monthly payment = £2,070 ÷ 24 = £86.25 [A1]

£ 86.25

[3 marks]

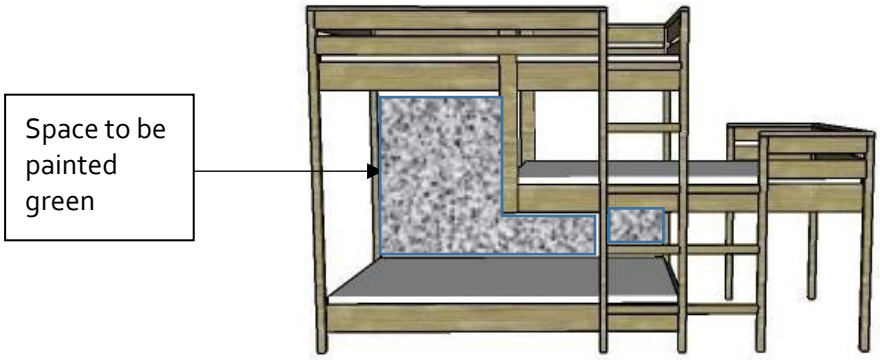
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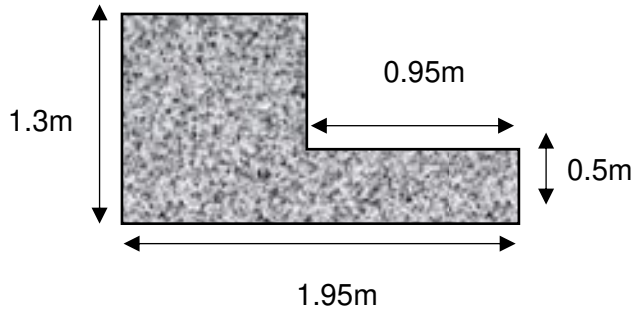
Q8

A man needs to paint the wall behind the bunk bed with a different colour next to each level of the bed.

The lower level will be painted green.



He needs to work out the area of this space.



Sketch of wall area
Not to scale

What is the area that he will paint green?

Show your working

Step 1: Split wall into two rectangles (Compound Shape):

- *Rect A: 1.3 m × (1.95 m − 0.95 m) = 1.3 m × 1.0 m = 1.300 m² [M1]*
- *Rect B: 0.5 m × 0.95 m = 0.475 m² [M1]*

Step 2: Total Area painted green = 1.300 m² + 0.475 m² = 1.775 m² [A1]

Area painted green 1.775 **m²**

[3 marks] **(3 marks)**

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Q9

A group of friends play a game of cards.

They have already taken 9 cards from the pack.

A player wants to know the probability that the next card is ♠

Card type	Number of cards in the pack at the start	Tally of cards taken already
●	2	I
♦	4	II
♠	6	
♥	6	I
♣	4	IIII
★	2	I

What is the probability that the next card is a ♠?
Give your answer as a fraction in its simplest form.

Show your working

- Total cards in pack at start = 2 + 4 + 6 + 6 + 4 + 2 = 24 cards
- Cards taken already (tally) = 9 cards --> Cards left = 24 − 9 = 15 cards [M1]
- Spades taken already = 0 --> Spades left = 6 cards [M1]
- Probability next card is a Spade = 6/15 = 2/5 (simplest form) [A1]

2/5

[3 marks]

(3 marks)

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Q10

A joiner has a video channel on the internet where he shares tips on how to make items for the home.

He wants to organise his videos into three suitable groups by time.

1 minute		1 minute		12 minutes		2 minutes		3 minutes		14 minutes
2 minutes		8 minutes		14 minutes		6 minutes		1 minute		7 minutes
12 minutes		4 minutes		9 minutes		11 minutes		3 minutes		14 minutes

Show how he will organise his videos.
Make one comment about what the results show you.

Show your grouping

Time Length Group	Video Lengths (mins)	Frequency
0 – 5 minutes	1, 1, 1, 2, 2, 3, 3, 4	8
6 – 10 minutes	6, 7, 8, 9	4
11 – 15 minutes	11, 12, 12, 14, 14, 14	6

Total: 18 videos [M2, A1]

Comment

- Most videos (8 out of 18) are short videos lasting between 0 and 5 minutes, while the fewest videos (4 out of 18) last between 6 and 10 minutes. [A1]

[4 marks] **(4 marks)**

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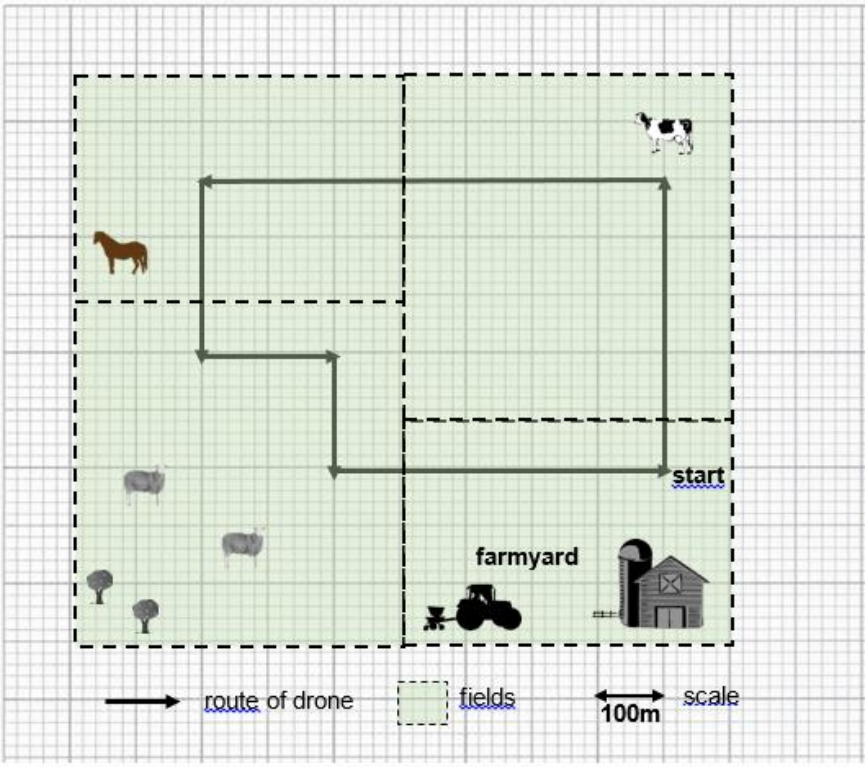
Q11

A farmer uses a drone to monitor animals.

The drone can fly a distance of 8km when its battery is fully charged.

The drone has $\frac{1}{4}$ of its battery life left.

The farmer needs to know if the drone has enough battery to complete this route shown starting from his farmyard.



Does the drone have enough battery life to complete the route?
Explain your answer with figures.

Show your working

Step 1: Drone remaining battery range = ¼ of 8 km = 8 km ÷ 4 = 2.0 km (2,000 m) [M1]

Step 2: Distance of route on grid (each square = 100 m):

Count grid segments = 24 squares × 100 m = 2,400 m = 2.4 km [M1, A1]

Does the drone have enough battery life to complete the route?

(tick one box) **Yes** ☐ **No** ☑

Explanation

1. The drone does not have enough battery life because the route is 2.4 km (2,400 m), but it only has 2.0 km (2,000 m) of battery range remaining. It is 0.4 km short. [A1]

[4 marks]

(4 marks)

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Q12

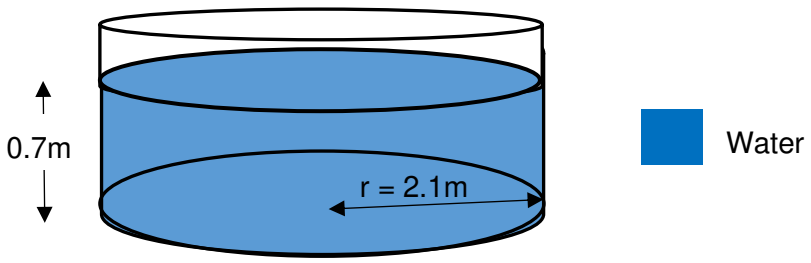
A man needs to work out what filter machine to buy for cleaning his circular swimming pool.

He uses the following formula to calculate the volume of water in his pool in litres

To find the number of litres

Step 1 Multiply the radius squared (r^2) by 3.142
Step 2 Multiply the answer to Step 1 by the height of water
Step 3 Multiply the answer to Step 2 by 1 000

Diagram of pool



The man would like a filter machine that can clean the full volume of water in his pool in less than five hours.

He finds a filter machine that cleans 2000 litres of water per hour.

He needs to know if this machine will be suitable for his pool.

Will this filter machine be suitable?
Explain your answer using figures.

Show your working

Step 1: Radius squared = 2.1² = 4.41 --> 4.41 × 3.142 = 13.85622 m² [M1]

Step 2: Multiply by water height = 13.85622 × 0.7 m = 9.699354 m³ [M1]

Step 3: Convert to litres = 9.699354 × 1,000 = 9,699.35 litres in pool [A1]

Step 4: Machine cleans 2,000 l/hr × 5 hrs = 10,000 litres capacity in 5 hours [M1]

Will this filter machine be suitable?

(tick one box) **Yes** ☑ **No** ☐

Explanation

1. Yes, the filter machine is suitable because the pool contains 9,699.35 litres,
and in 5 hours the machine cleans 10,000 litres (takes 4.85 hours < 5 hours). [A1]

[5 marks]

(5 marks)

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Q13

A man has a flight at 19:15 on Thursday. He must arrive at the airport 2 hours before this time to check in.

He needs to know what time he should leave home.

He will park his car at the airport car park and it will take 15 minutes to get to the check in counter.

The distance from his house to the airport is 90 miles.

He will drive at an average of 60miles per hour.

What time must he leave home?

Show your working.

Step 1: Flight time = 19:15. Arrive at airport 2 hours before:

19:15 – 2 hours = 17:15 at check-in counter [M1]

Step 2: Allow 15 mins from car park to check-in:

17:15 – 15 minutes = 17:00 at airport car park [M1]

Step 3: Calculate driving time (90 miles at average 60 mph):

90 miles ÷ 60 mph = 1.5 hours = 1 hour 30 minutes [M1]

Step 4: Time to leave home = 17:00 – 1 hour 30 minutes = 15:30 (or 3:30 pm) [A2]

Time to leave home 15:30

[5 marks] **(5 marks)**

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Q14

A woman receives an offer to join the Pet Club at her Vets.

> **Join the Pet Club!**
>
> For just £19.99 a month, we can offer the following exclusive benefits to help keep your pet happy and healthy:
> - FREE Annual Injection
> - FREE Flea & Worm Treatment
> - 50% off Health Check

She needs to pay the following costs for her dog

- £60 for the annual injection
- £42 for flea & worm treatment every three months
- £36 for a health check every six months

Should she join the Pet Club?
Explain your answer. Include figures to support your explanation.

Show your working.

- Option 1 (Club): 12 × £19.99 = £239.88 + checks (2 × £18 = £36) = £275.88 [M2, A1]
- Option 2 (No Club): £60 + Flea (4 × £42 = £168) + Checks (2 × £36 = £72) = £300.00 [M2, A1]

Should she join the Pet Club?

(tick one box) **Yes** ☑ **No** ☐

Explanation

1. Yes, she should join the Pet Club because the total cost for the year is £275.88, compared to £300.00 paying separately without the club.
2. Joining the club saves £24.12 over the year (£300.00 – £275.88 = £24.12 saving). [A1]

[6 marks]

(6 marks)

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Q15

A principal needs to report to the governors how absence rates have changed over the last four years.

She needs to work out last year's percentage absence.

College absences	
Total absences (days)	**Total possible attendance (days)**
948	39500

What percentage of the total days were the students absent for last year?

Space for working

- Percentage Absence = (Total Absences ÷ Total Possible Attendance) × 100
- Calculation = (948 ÷ 39,500) × 100 [M1]
- = 0.024 × 100 = 2.4% [A1]

Percentage absence 2.4 **%**

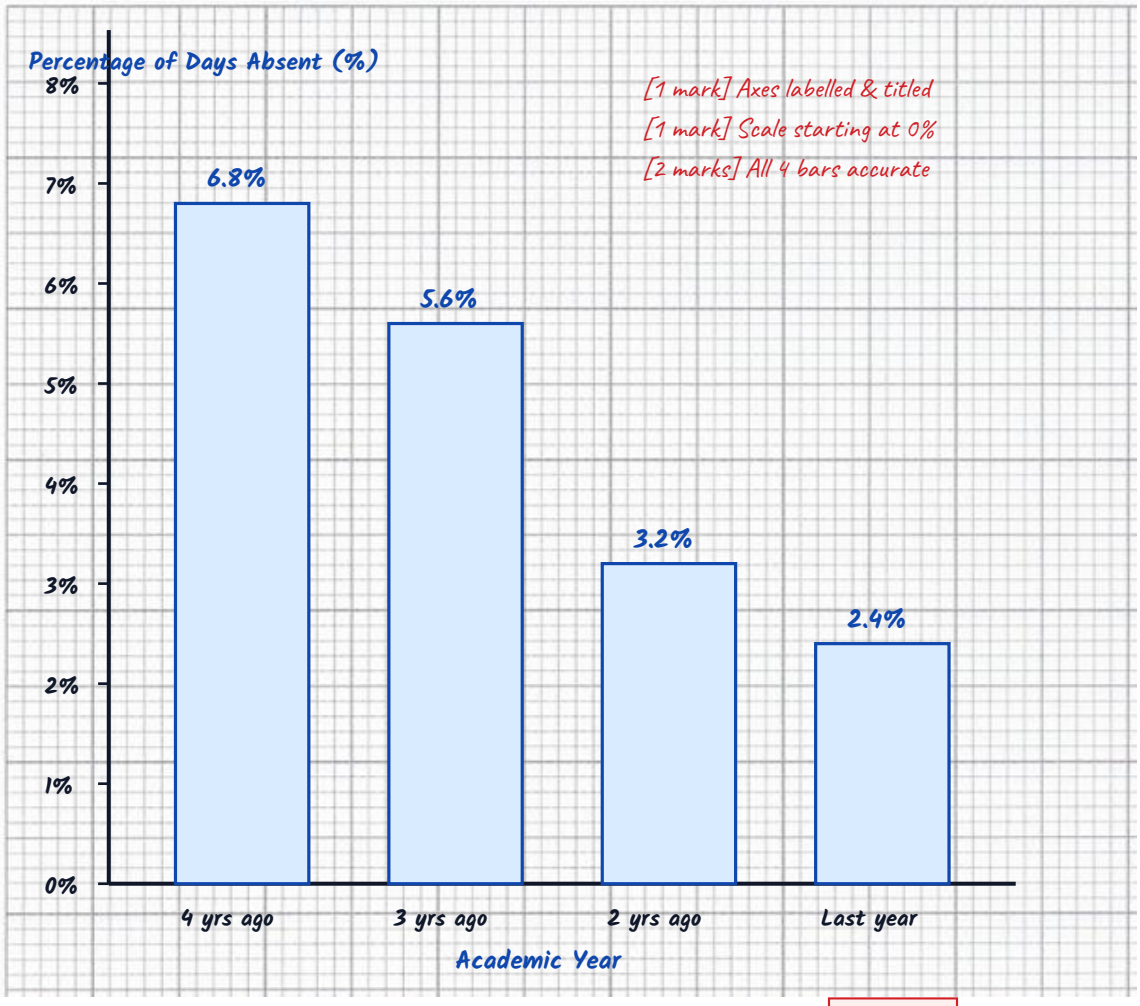
She wants a chart for her report.

She finds the following extra information.

	Percentage of days absent
4 years ago	6.8%
3 years ago	5.6%
2 years ago	3.2%

Draw a chart or graph for the principal's report.

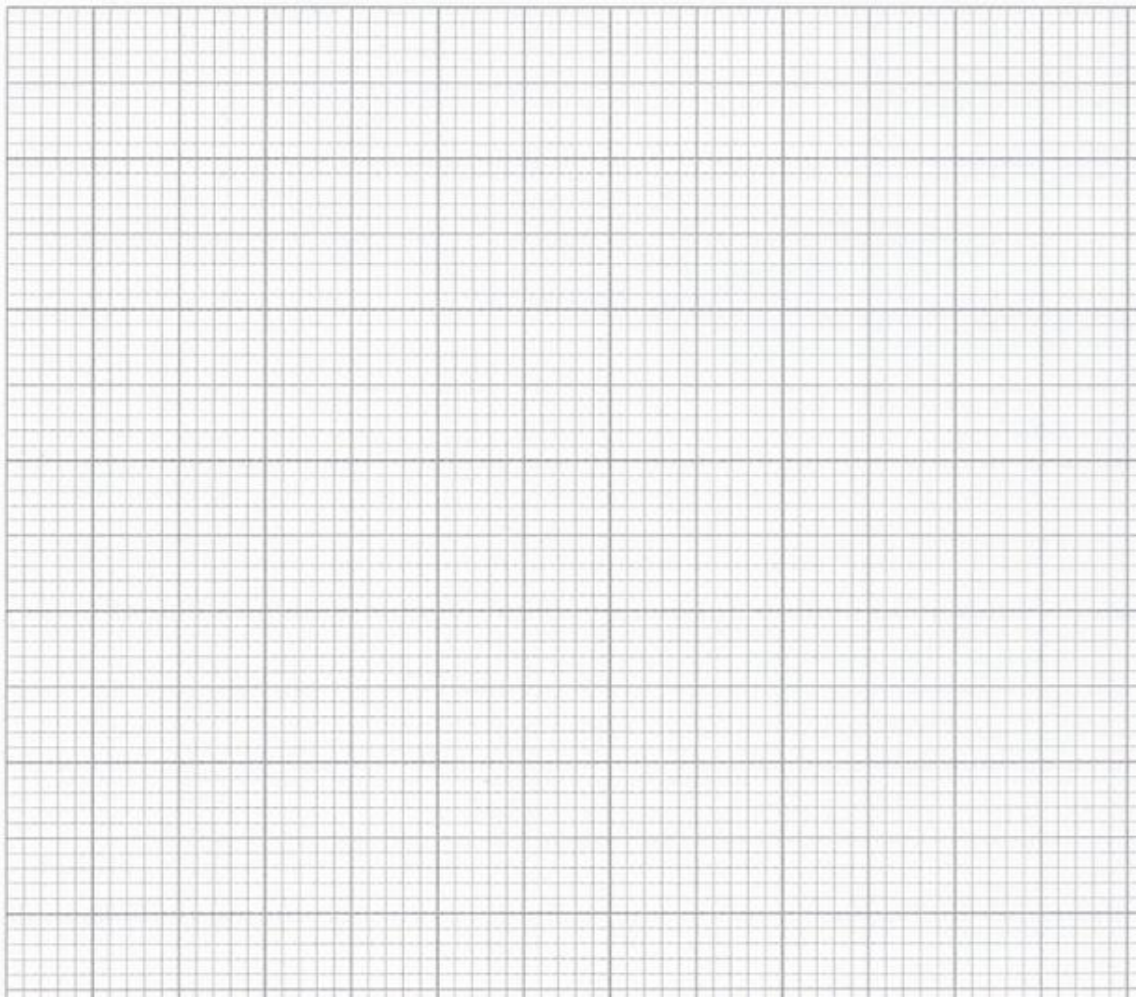
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(6 marks)

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Spare graph paper for Question 15



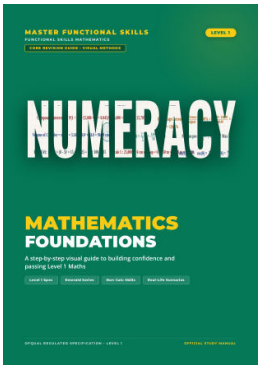
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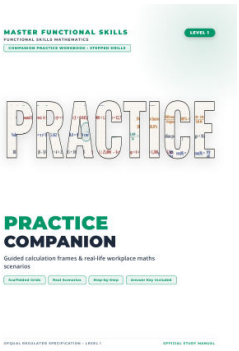


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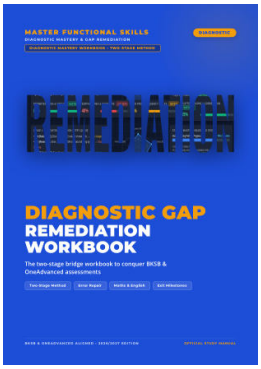


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